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InGen Physical AI Capstone Analytics

Market Intelligence and Platform Recommendations

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Key Findings and Recommendations

Operational Performance

- All three primary KPI means fall below proposed targets in the simulation.
- Recommendation: Use consistent KPI definitions and review monthly performance against targets.

Caregiver Support

- Caregiver support is the dominant topic in 12 of 20 articles.
- Recommendation: Test Fari's available functions with older adults and caregivers to validate user value.

Deployment Economics

- In one scenario, increasing the fleet from one to two robots reduces NPV from $2.9k to −$17.7k.
- Recommendation: Verify site costs and realizable labor savings before expanding the fleet.

Sources: Weeks 4–7 analyses. Operational and ROI results use synthetic data. Text findings reflect a selected 20-article corpus.



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Data Sources and Analytical Scope

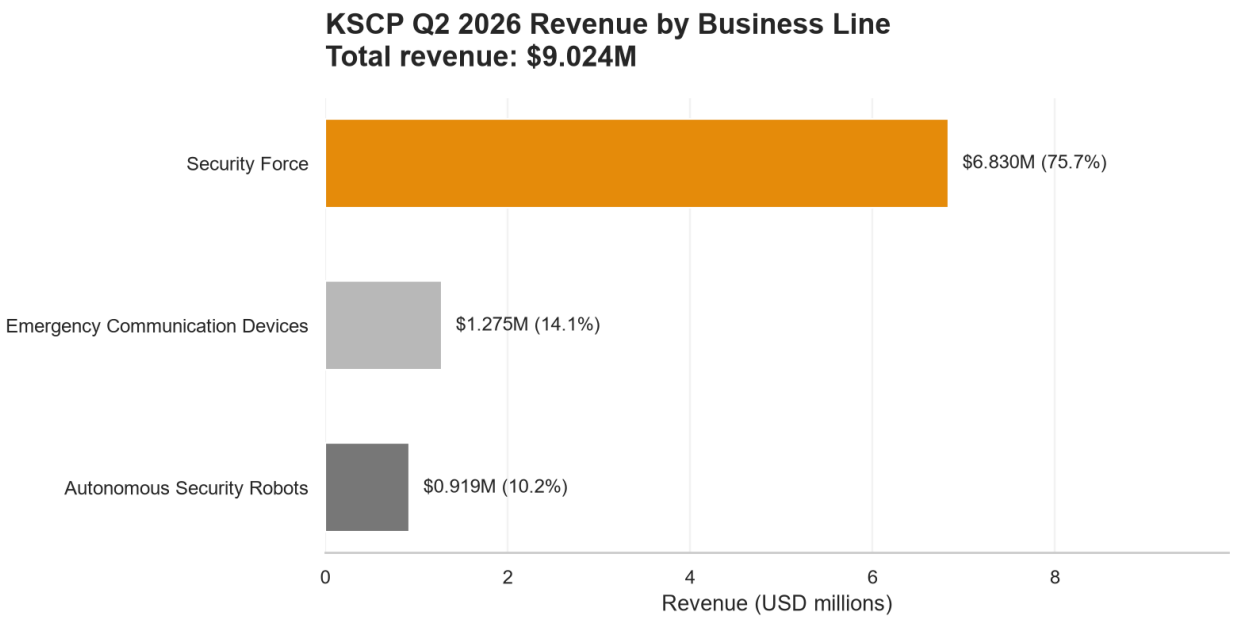
Focus: Aido Rover, Fari and Sentinel Prime AI

Dataset	Coverage	Analytical Purpose
Financial and wage data	SEC filings and BLS wage tables	Competitive analysis and cost benchmarking
Simulated operational data	180 monthly records across 3 platforms, covering 15 platform-specific KPIs	Performance analysis and Tableau dashboards
Product and industry articles	20 articles from 8 publishers	Topic modeling, sentiment analysis and exploratory classification
Simulated deployment scenarios	250 Aido Rover deployment scenarios	Three-year NPV modeling and sensitivity analysis

Sources: Weeks 2–7 project materials. Operational records and deployment scenarios are synthetic.

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Security Force Leads KSCP’s Q2 2026 Revenue



Source: Knightscope Q2 2026 Form 10-Q. Amounts in USD millions.

Total revenue: $9.024 million

Key Findings

- Security Force contributed 75.7% of total revenue.
- Autonomous Security Robots contributed $0.919 million, or 10.2%.
- Consolidated revenue therefore provides a limited measure of robot demand.

Implication for InGen

Track revenue and direct costs separately for robot deployments, software and human support.

Sources: Week 2 analysis and Knightscope Q2 2026 Form 10-Q.

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Simulated KPI Means Fall Below Proposed Targets

Platform	Primary KPI	Mean	Proposed Target
Aido Rover	Patrol coverage	93.3%	≥95%
Sentinel Prime AI	Incident detection	92.5%	≥95%
Fari	Active-day engagement	77.0%	≥80%

Key Finding

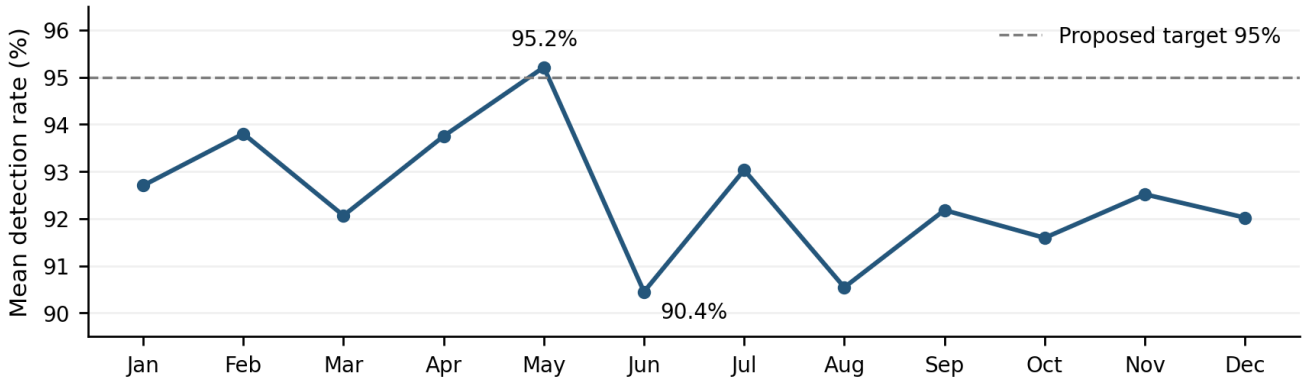
Outlier checks flagged no records in these three primary KPIs, although all three means were below their proposed targets.

Implication for InGen

Combine target comparisons with outlier checks to identify performance gaps.

Sources: Week 3 KPI Framework and Week 4 analyses. Synthetic data. Means equally weight 60 monthly records per platform. Targets are proposed benchmarks.

Sentinel Detection Varies Across Simulated Months



Key Finding

The monthly mean detection rate fell from 95.2% in May to 90.4% in June, a decline of approximately 4.8 percentage points.

The proposed target is 95%.

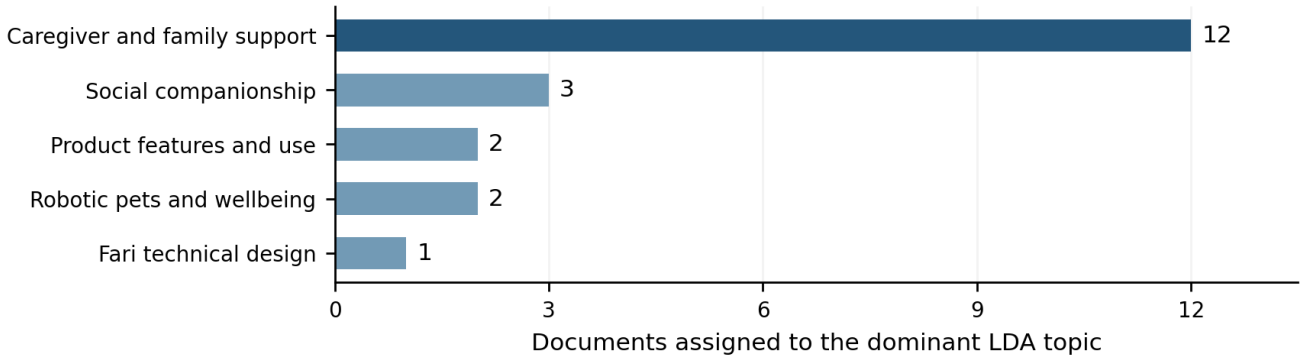
Implication for InGen

Monthly monitoring helps identify periods that need investigation. Review verified incidents and operating conditions when detection falls below target.

Sources: Week 3 KPI Framework and Week 4 analyses. Synthetic data. Monthly means across five simulated units. The variation does not establish a seasonal pattern.

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Caregiver Support Dominates the Collected Content



Key Finding

Caregiver and Family Support is the dominant topic in 12 of 20 articles, representing 60% of the corpus.

Data Limitation

Company content accounts for 55% of the corpus, limiting how broadly these findings can be generalized.

Implication for InGen

Use these themes to guide Fari pilot questions about daily engagement, reminders and caregiver updates. Validate their value with users.

Sources: Week 5 NLP Pipeline and Week 6 analyses. Exploratory findings from a selected 20-article corpus.

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Article Sentiment Can Differ from Product Evaluation

Article	Overall Product Assessment	VADER Result
The Senior List: ElliQ Review	Critical overall	Positive (+0.9986)
WHYY: AI Companionship Story	Generally favorable, with caveats	Negative (−0.6316)

Key Finding

Full-article sentiment combines product comments with emotional context, including loneliness and bereavement.

Other Exploratory Results

- Decision Tree matched 1 of 3 rule-based test labels.
- K-Means silhouette score was 0.0558, indicating weak cluster separation.

Implication for InGen

Evaluate product-specific statements and improve annotation quality before drawing conclusions about customer satisfaction.

Sources: Week 6 Sentiment Classification and Clustering Analysis. Selected article examples illustrate method limitations.

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Gradient Boosting Performs Best on Simulated NPV Targets

Model Comparison
Five-fold cross-validation on 200 training scenarios

Model	Mean CV RMSE (USD)	Mean CV R^2
Gradient Boosting	$28,357.41	0.9403
Linear Regression	$39,568.64	0.8843
Random Forest	$41,057.68	0.8768

Final Holdout Results

After model selection, Gradient Boosting achieved an RMSE of $24,205.13 and R^2 of 0.9589 on 50 test scenarios.

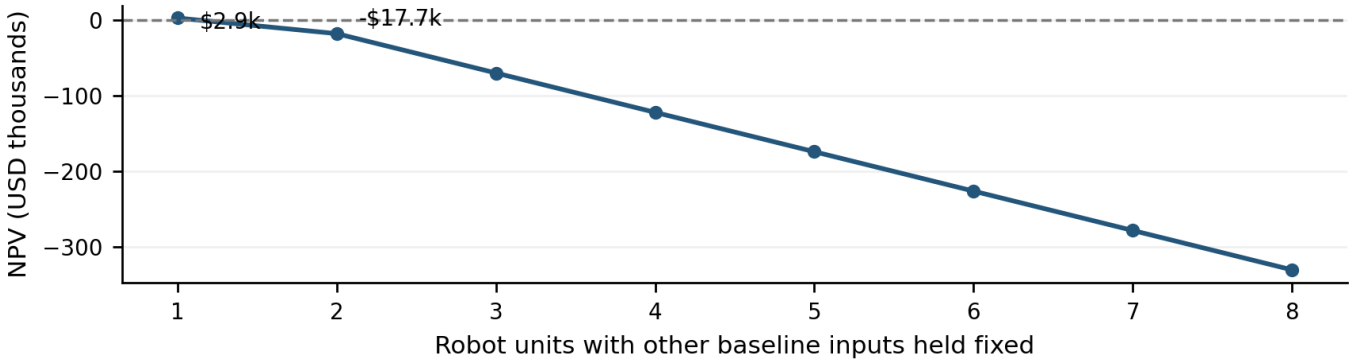
Interpretation

The model closely approximates the assumptions used to generate the simulated scenarios. Actual deployment returns still require independent validation.

Sources: Week 7 Predictive Model and Model Card. Synthetic deployment scenarios.

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Additional Robots Reduce NPV in the Selected Scenario



Key Finding

Increasing the fleet from one to two robots reduces calculated three-year NPV from $2.9k to −$17.7k.

Why It Happens

The second robot improves coverage, but its additional costs exceed the additional labor savings.

Beyond two robots, coverage is complete and modeled labor savings stop increasing.

Implication for InGen

Match fleet size to site coverage and staffing needs. Verify realizable labor savings before expanding deployments.

Source: Week 7 sensitivity analysis. Synthetic scenario with 4 staff FTE, 7.62 hectares and Standard subscription. Three-year NPV at a 10% discount rate. Other inputs remain fixed.

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Recommendations for Each Platform

Platform	Recommended Action	What to Measure
Aido Rover	Validate site economics before expanding the fleet	Realized labor savings, operating costs and uptime
Fari	Run a consented feasibility pilot with older adults and caregivers using available functions	Task completion, continued use and caregiver effort
Sentinel Prime AI	Link detection to verified incidents and response outcomes	Detection rate, false-alert share, response time and unresolved alerts

Shared Priority

Collect deployment data and evaluate results across independent sites or time periods before wider rollout.

Sources: Weeks 1–7 analyses and Capstone Report recommendations. These are proposed next steps.

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Evidence Limits and Next Steps

Evidence Limits

- Operational records and ROI scenarios are synthetic.
- The 20-article corpus is small and uneven across sources.
- Limited test samples and annotation quality restrict generalization.

Next Steps

- Collect real deployment data and verify financial assumptions.
- Expand the corpus across publishers and independently review labels.
- Evaluate performance on independent sites or time periods.

Closing Message

The project establishes a reusable analytics workflow. More representative real data and independent validation can strengthen future findings.

Source: Capstone Report, Methodology and Evidence Limits.

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